



# Translating non-experts' perception for expert engineers: A first step in co-designing automotive human-machine interfaces

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## ABSTRACT

This paper focuses on the study of a product experience's perception. It is then discussed how automotive Human-Machine Interfaces, as interactive products, are singular in their design process, perception, and evaluation. As essential products for a satisfactory user onboard experience, it is necessary for experts to have access to a translation of the non-expert perception, in a research and development perspective. Arguments about the benefits of using a sensory evaluation method, adapted to this kind of complex product, are provided. In this study, a sorting task evaluation is performed on naive people, to quantify and qualify the perception of a large number of stimuli. The paper proposes a statistical analysis methodology for processing data from a complex product evaluation. The analysis offers several levels of perception interpretation in order to provide a translation at different levels as well. An application example is given for three attributes that are different in nature and interpretation.

## 1. Introduction

In automotive context, Human-Machine Interfaces are onboard hardware and software allowing control and communication with the car, such as: panel instrument, central screen, air conditioning buttons, steering wheel controls, etc. (Meixner & Müller, 2017). These automotive HMI<sup>1</sup> are constantly evolving, numerous, very different from one to another and contribute to the construction of the cockpit. Furthermore, as interactive products, HMI are decisive for the User onboard Experience (UX) (Lallemand & Gronier, 2018). Ergonomic and hedonic researchers always sought to improve that experience by developing rules for the usability conception of interfaces (efficacy, efficiency, satisfaction) (ISO 9241-11, 2018). They have also developed models to explain that hedonic features, such as stimulation, identification, and evocation, are as important as ergonomic features. Combined, hedonic and ergonomic features are sources of pleasure and emotional reactions that contribute to satisfaction (Hancock et al., 2005; Hassenzahl, 2005; Thüring & Mahlke, 2007). HMI is therefore a particularly important component of the car for the manufacturer.

Each manufacturer carefully designs HMI in order to improve them, for a successful and secure onboard experience. In practice, many teams

are working on this task, such as ergonomists, designers, IT specialists, and so on. The multiplicity of actors, sciences and measures used in the automotive HMI design, makes them complex. In other words, this is a complex product because of its multidimensionality in terms of:

- Components: HMI are characterized by a multiplicity of positions inside a car, as well as a multiplicity of functions such as driving assistance, comfort, or entertainment (Meixner & Müller, 2017). The user is not dealing with one HMI, but several at the same time. What is perceived is the result of many different products. It is like being in front of a buffet composed of several food products and thus perceiving different flavors and textures. The manufacturer's goal is to ensure that the assembly of all these components, within a particular cockpit, is well done: each HMI must be visible, accessible and the resulting overall perception must be coherent.
- Interactions: As technological products, HMI allow for a diversity of interaction levels (physical, tactile, vocal, visual, etc.). In a car it is now possible to handle the radio with the central touch screen or even with the physical steering wheel controls. In addition, interaction with one HMI most often leads to the response of another one. For example, the turn signal's control is not dissociable from the

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<sup>1</sup> Human-Machine Interfaces

screen that displays the indication activation. To make the analogy with food, it is as if eating a product on a buffet automatically has an impact on the appearance of another product. The manufacturer's goal is to offer HMI with effective and logical interaction.

- Sensory stimuli: Each interface (button, screen) is made of various materials with different sensory levels (color, shape, texture, etc.). As interactive products, each control or display is defined by a specific visual or haptic feedback (light intensity, force, vibration, sound). The manufacturer's goal is to provide pleasant and ergonomic stimuli.
- Ergonomic features: According to Hassenzahl (2005), "manipulation of the environment requires relevant functionality (i.e., utility) and ways to access this functionality (i.e., usability)" (Hassenzahl, 2005). The automotive context also imposes safety of use as a requirement. A food product offers almost the same feeling of food safety when eaten in static (sitting) or in dynamic (walking). Whereas, in the automotive context security takes priority because from the moment there is a dynamic setting (the vehicle is moving), the risk of making an accident or suffering one can be important. In this context, it adds a layer of complexity to the design because HMI must be easy to use but also provide safety. As driving is the driver's main task, in general, HMI should not alter the driver's attention. They should also be accessible, readable and understandable.
- Hedonic features: According to Hassenzahl (2005), "hedonic attributes emphasize individuals' psychological well-being". Hedonic features, such as stimulation (providing new impressions, opportunities, and insights), identification (expressing the self through physical object) or evocation (provoking memories) are essential to ensure pleasurable experiences (Hassenzahl, 2005). Regardless of the brand, if hedonic features are essential to the packaging of food products for example, the amount of sugar contained on it (the ingredient as such) has no impact on the image I project to myself and to others. On the contrary, in the case of automobiles the amount of chrome can have an impact on the image I project to others as the owner of the product. The hypothesis is that, from one culture to another, the image I send back thanks to this material can be the image of a successful person.

Each dimension is divided into sub-dimensions and each one interacts with the other. Many visual stimuli (lighting animation, navigational aid with three-dimensional modeling arrows, etc.) could negatively impact the driver's attention, while evoking a technological image. A weak haptic feedback, when manipulating a touch-sensitive button, could be pleasant, while negatively impacting the ergonomics if the driver has any doubts about whether their request has been taken into account by the HMI.

Although carefully constructed, Hassenzahl (2003) highlighted the fact that the designer's perspective was different from the user's one. In fact, there is no guarantee that the product will be perceived as intended. On one hand, non-expert perception is depending on various factors external to the HMI, i.e. context of use (e.g. driving in town or through the countryside does not have the same impact), but also user's habits, moods and expectations (Gil Urrutia et al., 2017; Hassenzahl & Tractinsky, 2006). On the other hand, because the product is inherently interactive, there are not only sensory dimensions to be considered but also ergonomic and hedonic ones. In the automotive context, it is necessary to consider ergonomic dimensions mainly because of the security aspect, and hedonic ones mainly because of the special bond that can be created between a user and his vehicle. Indeed, buying a car matters and the cumulative time spent in a car, 4 years in average for a European (Citroën France, 2016), is so important that it could be considered as a house extension. Thus, inside a cockpit, a user makes a mixture of a large number of sensory, ergonomic and hedonic stimuli and creates a personal and mental product through their own emotional experience before making an overall judgment (Hassenzahl, 2005).

How an interface, designed through a set of such heterogeneous

dimensions, can be perceived by its user, is the question addressed in this study. This question is essential for the manufacturer as it would allow him to understand the difference between his own perception and that of a non-expert. This is the first step in co-designing products.

Subsequently, objectives of this current study are (i) to bring a reflection on using a sensory methodology to understand non-expert perceptions of a complex product. Indeed, automotive HMI is a complex product mainly because of its multidimensional perception (ergonomic, sensory and hedonic dimensions are not separable) and its context dependence. Therefore, it is expected to (ii) explain how the sensory evaluation methodology employed gives access to a non-expert perception, by presenting how the data collected can be analyzed, to identify the complex perceptual elements. Finally, it is expected to explain (iii) how a non-expert perception can be interpreted to help experts in the quest for a translation of ergonomic and hedonic attributes. At the same time, (iiii) a reflection on the use of such a methodology in a particular evaluation context, that of the Covid19, will be done.

## 2. Method & material

### 2.1. Sensory methodology

It is expected to use an evaluation methodology that allows to quantify and qualify non-expert perceptions. However, HMI is complex to assess because of the interaction of various stimuli when the user is inside a cockpit. First, HMI are multisensorial because they have as many sensory features as the number of physical components that constitute them. Combined, components create a mixture of many sensory stimuli (Blumenthal, 2001). Indeed, each HMI is the result of several visually perceptible elements as opposed to an alimentary product where ingredients can be invisible for human eyes. Each element, for example the ones that constitute a central screen (on-off button, touchscreen, display, frame), is defined by a specific combination of sensory levels: visual, touch, and sound. Then, automotive HMI are implemented in a dashboard. It seems that the environment (i.e. dashboard) is non-dissociable from user perception. Herberth and Blumenthal evaluated the impact of the dashboard design on the instrument panel perception (i.e. screen placed behind the steering wheel) and highlighted that there was a close and positive relationship between hedonic perception of environment and hedonic perception of HMI (Herberth & Blumenthal, 2019). Finally, the context of use implies that the security component is increased because the action of driving remains the priority during HMI interaction, which gives less time for the user to focus on the onboard experience. In that context other stimuli related to the road are impacting the perception. Therefore, usability satisfaction is as much as important as the feeling of security, both having an impact on each other (Meixner & Müller, 2017). This complexity must therefore be considered in the choice of the evaluation methodology. Since HMI and environment are indissociable, the evaluation of a large number of different products seems necessary, in order to vary cockpit dashboard. However, conducting an evaluation while driving seems inconceivable for safety reasons. Therefore, the non-expert will be asked to project himself driving.

The product studied here is interactive, moreover it is an automotive one. However, classical methodologies used in UX design do not seem to be suitable for several reasons. (i) When submitting psychometric questionnaires, such as the AttrakDiff developed by Hassenzahl and al. (2003) or the modular evaluation of Components of User Experience (meCUE) developed by Minge and Riedel (2013), methods are not specific to the automotive context. In addition, they are mainly focused on the global feeling assessment (is that easy to use? Is that pleasant to use?) without identifying elements involved. (ii) When conducting more specific interviews to qualify perceptions, methods are not adapted to simultaneous evaluation of several HMI or dashboards because it is time consuming. Indeed, interviews are more adapted to a monadic test

(Hassenzahl et al., 2003; Lallemand et al., 2015; Minge et al., 2016). To aim the objectives, a method from psychological sciences, but as well used in sensory sciences as in UX design, is considered: a sorting procedure. This method can be easily adapted to the automotive context, it allows a simultaneous evaluation of several HMI and is lower time costs.

UX design and sensory sciences seem to differ in both their objectives and measurements. In UX evaluation, the user will be asked about his own perception of the interaction with a product. This is a subjective evaluation. “The product would enhance my standing among peers” or “I consider the product extremely useful” are examples of questions asked in the meCUE UX questionnaire (Minge et al., 2016). If he is not questioned, the user will be observed, and his performance with the interactive product will be evaluated. In classical sensory evaluation, the panelists will be asked about the objective sensory characteristics of a product. Thus, in UX design several variables, often subjective, are measured to approach a latent variable that cannot be measured directly (ergonomic or hedonic quality). According to Alben (1996), it is “the way it feels in their hands, how well they understand how it works, how they feel about it while they’re using it, how well it serves their purposes, and how well it fits into the entire context in which they are using it” that will characterize the product (Alben, 1996). Whereas in sensory evaluation, directly measurable variables are studied (e.g. sweetness intensity) while considering inter-individual differences. Then, a link with a global appreciation can be performed and will allow access to a more subjective evaluation of the product (evaluation of the preference). Despite the differences, sensory sciences and UX design both have a functional background with their own set of evaluation instruments. Respectively, the study of organoleptic qualities with sensory analysis, the study of pragmatic qualities with user tests thanks to usability scales. The two disciplines meet when the objectives converge. It probably occurs when the needs of understanding the product are consumer oriented. It is then observed that new kinds of sensory evaluation, more experiential, are designed. Indeed, authors are increasingly taking into consideration the context or the hedonic perception such as emotions in their studies, although standards recommend standardization of the evaluation environment and objectification of the measurements. For example, in 2019, Adriana Galiñanes Plaza studied the effects of context on consumer hedonic evaluation of food products (Galiñanes plaza, 2019; Martins et al., 2021; Porcherot et al., 2022; Sinesio et al., 2021). When objectives converge, no methods should be neglected regardless of their origin. In hedonic evaluation, methods can be more transversal because they come from psychology (sorting tasks, emotional assessment tools, etc.). The sorting task is a fast and non-verbal method of assessing the similarity of the stimuli under study (Depledge & Société scientifique d’hygiène alimentaire, 2009). Already used in the 20th century for the understanding of emotions (Hulin & Katz, 1935), they are now usually applied in UX design and sensory sciences. In UX it is used to understand mental patterns in order to improve the architecture of systems (Lallemand & Gronier, 2018). In sensory sciences, it is used to understand the connection between products. In this paper, the sensory based approach has been chosen to answer the objectives of understanding the similarity between products considering specific attributes. It is then expected that verbalization will provide access to ergonomic and hedonic perceptions and allow to combine quantitative and qualitative data collection from non-expert participants.

Participants were requested to sort sixteen photos of cockpits (i.e. stimuli), for three different attributes borrowed from HMI experts in Renault Group. The sorting task was not free. Three categories were imposed each time to reduce the cognitive effort from participants, but also to ensure the confrontation of conceptual perception from experts and participants, i.e. how both interpret an attribute within the same stimuli set (Brard, 2019). Indeed, experts’ conceptual perception can easily be summarized on a three-point scale: high and low levels (i.e. meets or does not meet the concept) and an intermediate one. Then participants were asked to verbalize their categorizations in order to obtain information about which specific criteria impacted their choices.

## 2.2. Attributes

Three conditions were followed to select attributes from Renault’s expert reference framework. Attributes must imply the sight, offer various difficulties in understanding, and refer to hedonic and ergonomic perception. Finally, the three following attributes were evaluated: Visual Load and Screen Integration that refer to a hedonic dimension; Interface Usability that refers to an ergonomic dimension. This presentation order was purposely chosen to mimic the natural experiential process, from discovery to use. Visual Load refers to HMI first impression, Screen Integration refers to aesthetic, then Interface Usability refers to ease of handling. The expert translations, for those attributes, are given in Table 1. This table will be used as a reference to compare the expert translation with the non-expert one at the end.

## 2.3. Stimuli

The aim was to obtain a set of 16 stimuli that was as representative as possible of the existing diversity for Visual Load, Screen Integration, and Interface Usability attributes. Within the online evaluation setup (see 2.4.), stimuli were presented as photos. A suitable set of pictures was selected from publicly available pictures on the Internet (e.g. www.netcarshow.com), with appropriate angles, zooming and resolution, so that all necessary details were visible to participants. Concerning photo features, for the first two attributes (Visual Load and Screen Integration) two angles were presented, a front and a right-side view (a left-side view being rarely found). For the last attribute (Interface Usability), a zoomed-in view of the considered interfaces was added. Concerning screen layout, for classic cockpits (i.e. cockpits with an on-board display, a central screen and physical buttons for the main comfort and entertainment functions), only the homepage was presented on screens. For more unusual cockpits (i.e. cockpits with very large screens that handle all functionalities of driving assistance, comfort, and entertainment), homepage, or a combination of different menus, or else air conditioning menus were displayed, depending on availability. About environmental standardization, some elements of the cockpit were virtually removed from the photos: seats, outside elements visible through windscreen or windows, brands located on steering wheel or screen. These modifications have been realized with Paint 3D (Klaxoon, 2021) (6.2105). A stimuli example is given in Fig. 1.

## 2.4. Protocol structure

To ensure the correct quality of the qualitative data collection, but also to meet the requirements of the Covid19 sanitary situation, the study was set up on the online platform Klaxoon (Klaxoon, 2021), which is a collaborative platform for interactive workshops. Four activities were planned: a socio-demographic questionnaire followed by three sorting tasks. Each set of instructions was given aloud, and 16 stimuli were presented on virtual sticky notes, randomly numbered with 3 digits. Participants were able to manipulate each sticky note: moving it, zooming-in on the photos. After having explained the task at hand (see Table 2), we gave the participants all the time needed to sort all the stimuli. Each imposed category was presented as a colored column and was qualified by a specific level (see Table 1). They just had to drag the sticky note to the desired category, which made it change color and level. Then, each group of stimuli formed was presented again but separately to the participants, and they were asked to verbally explain why they did choose to put all the stimuli considered in the selected category. The study lasted about 60 to 90 min per participant. At the end, two types of data were collected: for each attribute and each participant, stimuli qualified by levels and a verbalization of categories.

## 2.5. Participants

Forty people agreed to participate in this study. Twenty-five women

**Table 1**  
Attributes selected for the study.

| Attributes          | Definitions                                                                                                                       | Intended understanding                                                                                                                                                                 | Levels                                                              |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Visual Load         | Number of information returned by the dashboard, which is associated with the number of HMI inputs (commands, screens) presented. | An easy-to-understand attribute because HMI are enumerable, so dashboards are directly comparable between them. The more HMI signals, or information, the heavier the load.            | Loaded<br>Intermediate<br>Pure                                      |
| Screen Integration  | 1. Embedding in the dashboard.<br>2. Between-screen and screen-interior harmony.                                                  | Less easy-to-understand attribute because integration is not enumerable. Screen comparisons involve different criteria (position, shape, color, direction, thinness, size, gap, etc.). | Successful integration<br>Incomplete integration<br>Bad integration |
| Interface Usability | Ease to understand and to operate.                                                                                                | Involves a projection into the interaction which can be difficult to perform but stimuli remain visual, and the attribute is easy to understand.                                       | Simple<br>Intermediate<br>Complex                                   |



**Fig. 1.** Stimulus example.

**Table 2**  
Given instructions.

| Sorting task | Sort these photos according to how you feel about: |
|--------------|----------------------------------------------------|
| 1            | The perceived visual sensory load.                 |
| 2            | The integration of screens in the cockpit.         |
| 3            | The projected ease of use of the interfaces.       |

and fifteen men of 18–60 years old, having a driving license, took part in the study. The experience feedback about the online tool was positive, participants finding it easy to handle. The distance and possible connection issues were not an obstacle to the smooth running of the evaluation.

## 2.6. Data analysis

The methodology proposed consists in the search of indicators allowing to interpret how the attributes have been perceived. The analysis will be done in a sequential manner, and the consensus on participants' responses will be analyzed from different points of view. Each attribute will be investigated through the perception of the stimuli, the use of the categories that constitute the attribute, and their qualitative descriptions. These different steps should provide a translation of a non-expert conceptual perception, that could be invaluable for the expert engineers. The different analyses were carried out using R (4.0.5) and the FactoMineR package (Lê et al., 2008).

The first step consists in performing a statistical analysis on the stimuli classification. To evaluate the consensual nature of the attribute perception, data were structured with stimuli (i.e. products) as statistical individuals, and participants' answers as variables (Cadoret et al., 2009). A two-stage analysis is then performed on the resulting data matrix: A Multiple Correspondence Analysis (MCA), followed by a Hierarchical Ascendant Clustering (HAC). The HAC, carried out on

observation coordinates in the MCA solution space, is consolidated with a K-means method. To test for the significance of the induced data structure, intra-column permutations were performed to simulate the distribution of the first eigenvalues under the null hypothesis, i.e. the perception of the attribute from the point of view of the stimuli is not consensual. If the stimuli classification is consensual between participants, then products perceived as similar should be distributed in separate clusters. Then, the consistency of the stimuli qualification is analyzed. If the concept perception is consensual between participants from the point of view of the modalities, then each cluster should be most often qualified by the same attribute level.

The final step consists in performing a statistical analysis on the verbalizations of categories. To evaluate how concepts behind the attribute are understood by participants, a semantic analysis, based on the hypergeometric distribution was performed on their verbalizations (Husson et al., 2017), in order to highlight the terms most frequently used. By aggregating the descriptive sentences with similar meanings, we obtain a textual interpretation of each attribute level.

To interpret the attribute perception, stimuli classification, concepts perception, qualitative descriptions and a visual description from photos were associated.

## 3. Results

A detailed use of the methodology is made for the first attribute, but only the main results will be presented for the others.

### 3.1. Visual load

At the stimuli level, there is no ambiguity observed. According to the permutation tests, data are significantly structured. An important and significant part of the information is explained by the first two factorial axes. In addition, the analysis of the eigenvalues (see Fig. 2) shows that there is no interest in adding an additional dimension. Also, three homogeneous clusters are identified (see Fig. 3) and are well distributed on the two first factorial axes (see Fig. 4). Therefore, stimuli perception appears reasonably consensual for the Visual Load attribute, especially for the paragon (i.e. individuals located in the centers of the clusters), considered as unambiguous representatives of each cluster. For some stimuli located at their cluster boundaries, ambiguity may be slightly higher (see stimuli 213, 835 and 457 from clusters 1 and 2 on the Fig. 4).

At the concept level, there is no ambiguity observed because, for each cluster, stimuli qualification is consistent among participants. Hence, cluster 1 gathers products that have been put in the loaded category, cluster 2 gathers the intermediate ones and cluster 3 gathers the purified ones. Therefore, the concept perception also appears to be reasonably consensual for the Visual Load attribute. Pure seems to be the most distinguishable concept since the cluster is the farthest apart from the other two.

Table 3 lists the sentences that are over-represented in the verbalization of categories. Considering the 40 participants, the sentences with

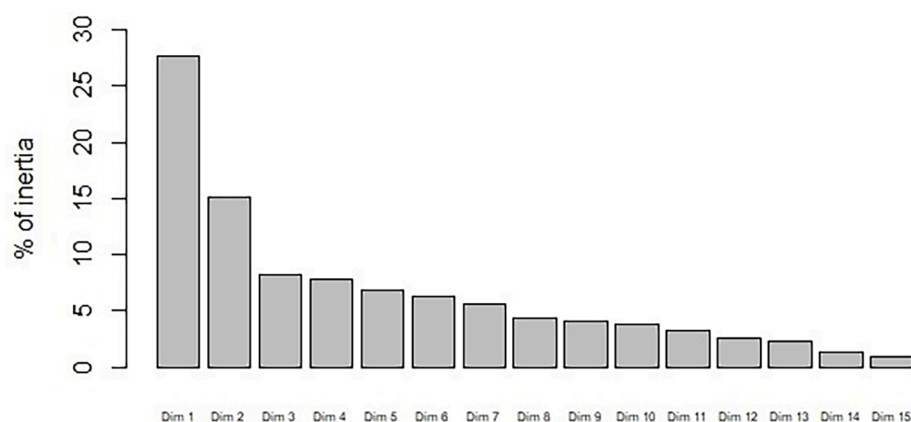


Fig. 2. Bar chart of eigenvalues from the MCA for the Visual Load attribute.

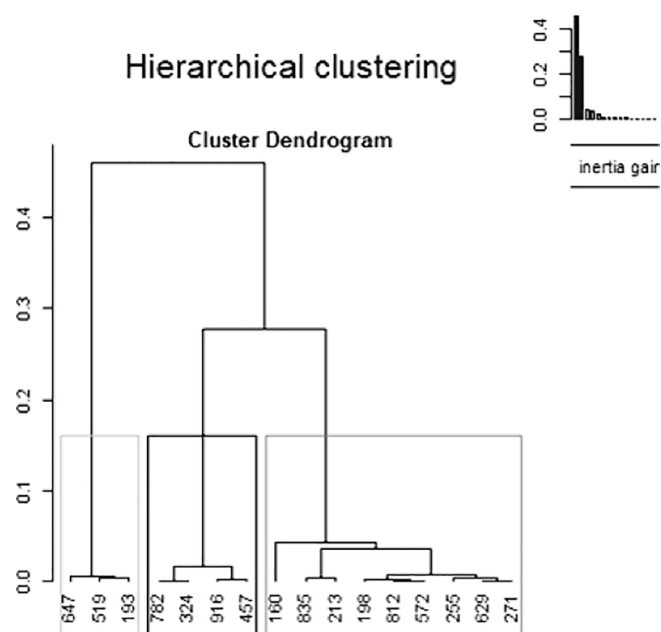


Fig. 3. MCA + HAC + K-means results as a dendrogram for the Visual Load attribute.

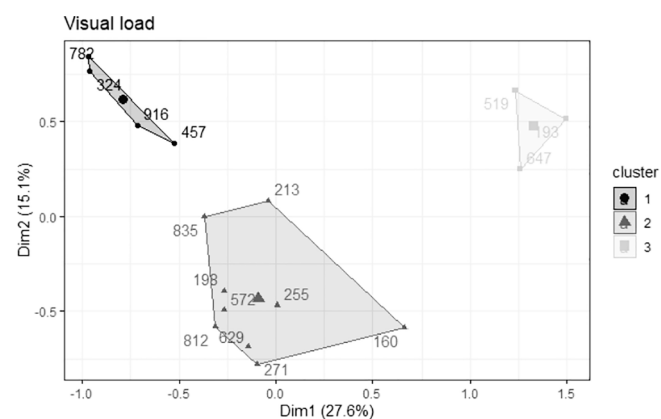


Fig. 4. MCA + HAC + K-means results in the first 1x2 factor plane for the Visual Load attribute.

Table 3

Semantic analysis results for the Visual Load attribute.

|                                                                           | p-value |
|---------------------------------------------------------------------------|---------|
| <b>Load</b>                                                               |         |
| Too many buttons                                                          | 0.00    |
| Too many different screens                                                | 0.00    |
| At the design level many details like floors and colors                   | 0.01    |
| Lots of buttons on the steering wheel                                     | 0.01    |
| Too much information                                                      | 0.02    |
| <b>Intermediate</b>                                                       |         |
| Fewer buttons                                                             | 0.00    |
| Gearshift lever not appearing loaded                                      | 0.00    |
| Smaller screens                                                           | 0.01    |
| Less on the steering wheel                                                | 0.13    |
| <b>Pure</b>                                                               |         |
| The more it is organized into section on a screen the more it is purified | 0.00    |
| The lines are straight and aerial                                         | 0.00    |
| Just a screen that seems to fit the whole car                             | 0.01    |
| No buttons at all                                                         | 0.01    |
| Even the air vents are integrated into the lines                          | 0.07    |

a critical probability lower than 0.15 have been kept. According to the semantic analysis, two main criteria were used by the participants to classify stimuli: the HMI criterion and the design one. For the Loaded and Intermediate categories, different HMI were considered: physical controls, screens, steering wheel, gear lever. The more of them or the more buttons on them, the more loaded the stimulus was considered. Concerning design, dashboard colors and reliefs were considered. The more color and relief there is, the more loaded the stimulus. For the Pure category, two HMI were considered: the central screen and the physical buttons. If the screen gathers all the dashboard buttons, then the stimulus is considered as pure. Concerning the design, mainly the lines induced by the dashboard were considered. The straighter the lines, the purer the stimulus.

Regarding paragon, the visual description of the photos shows that the cockpits from the Load cluster (stimuli 916 and 324) are characterized by either multiple physical buttons (on the dashboard or on the steering wheel) and by a small central screen, or by 2 central screens on which multiple touch buttons are displayed. For the Intermediate cluster (stimuli 255, 198 and 572), cockpits are characterized by small screens and few buttons. For the Pure cluster (stimuli 193, 519 and 647), cockpits are characterized by a larger screen, almost no buttons, and by continuous lines and no visible air vents. There is consistency between the visual description of the unambiguous representatives and the textual analysis.

Regarding the ambiguous stimuli of the Intermediate cluster, cockpit 213 seems visually loaded, because of its multiple screens and its loaded steering wheel. It may be that it was considered as an intermediary stimulus because of its dashboard, as the criteria of the pure category is

respected (i.e. “lines are straight and aerial”, “even the air vents are integrated into the lines”). Cockpit 835 seems visually loaded because of its dashboard (a lot of relief and apparent air vents) and its numerous buttons on the steering wheel. But there are few buttons on the central part and there is a small screen, which may explain why it was considered an intermediary. Regarding the ambiguous products of the Load cluster, cockpit 457 seems visually loaded because of its multiple screens. But 457 comes close to the Intermediate category perhaps because of the straight lines, accentuated by the wood and the screen set up, and the few buttons visible on the central part.

### 3.2. Screen Integration

Here, there is ambiguity at the stimuli perception level and at the concept perception one. Indeed, the three clusters formed are not homogeneous and they need to be spread over more MCA dimensions to be explained (see Figs. 5 and 6). Thus, stimuli are not perceived in a homogeneous way by the participants. In addition, each cluster is qualified by at least two attribute levels. Thus, concepts of Successful, Bad, and Incomplete integration are not perceived in the same way by the participants. The semantic analysis for each category (see Table 4), shows us a strong semantic multidimensionality. Size of the screen, position with respect to the driving, and screen-dashboard consistency are considered as criteria.

### 3.3. Interface usability

Here, there is no ambiguity at the stimuli perception level, the information is significantly structured, and the clusters formed are homogeneous (see Fig. 7), but there remains some ambiguity in the concept perception. Indeed, clusters are qualified by more than one level. Cluster 3 is characterized by a Complex, Intermediate and Simple projection in use at the same time. However, there is a certain coherence between clusters 1 and 2, respectively: Simple and Intermediate levels on one side, Complex and Partial levels on the other side. Stimuli perception being homogeneous, it is interesting to perform a descriptive analysis to give an interpretation of the identified ambiguity. According to the visual description: cluster 1 and 2 are represented by classical cockpits (i.e. cockpits with an on-board display, a central screen and physical buttons for the main comfort and entertainment functions), while cluster 3 is represented by unusual cockpits with very large screens that handle all the information. Difference between cluster 1 et 2 seems to be made in the amount of HMI and information. Semantic analysis seems to be consistent with the visual one, but a strong semantic multidimensionality is observed for each category (see Table 5). Several dimensions are considered in the sorting: projection in the driving, amount of HMI, screen organization, size and position of the screen, quality of the information, and format of the touch buttons.

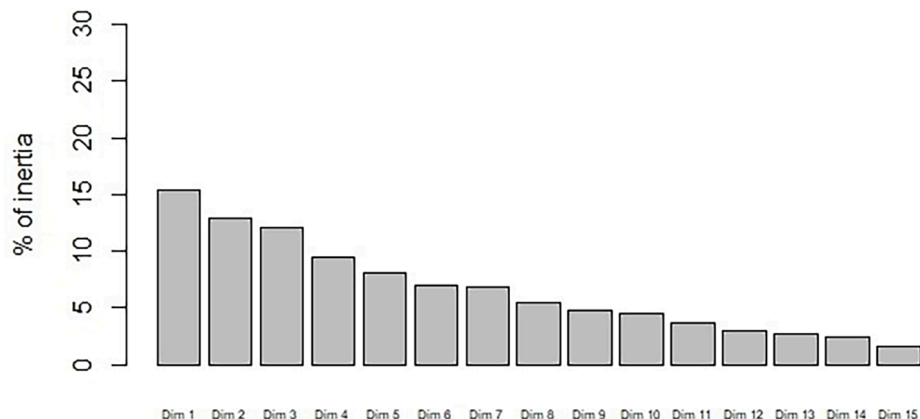


Fig. 5. Bar chart of eigenvalues from the MCA for the Screen Integration attribute.

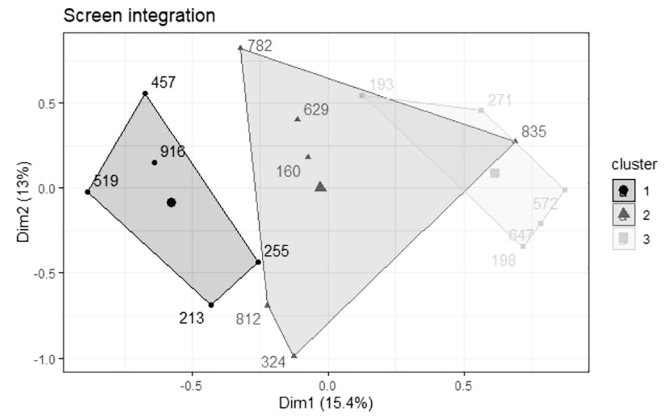


Fig. 6. MCA + HAC + K-means results in the first 1x2 factor plane for the Screen Integration attribute.

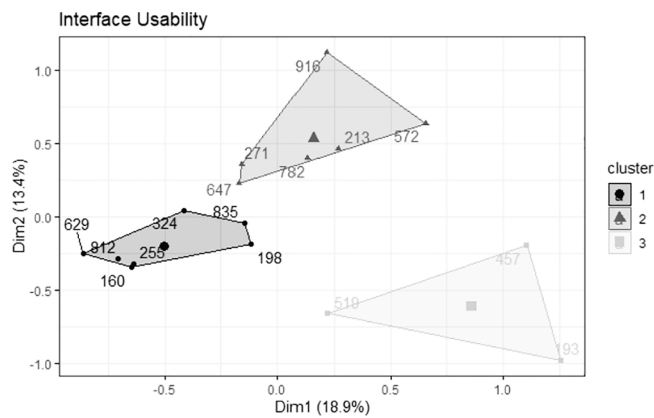
Table 4

Semantic analysis results for the Screen Integration attribute.

|                                                                                  | p-value |
|----------------------------------------------------------------------------------|---------|
| <b>Bad</b>                                                                       |         |
| Screen is a body annex of the car                                                | 0.00    |
| The driver has to lower his head a lot and it can impact the driving             | 0.07    |
| Too many screens to focus on                                                     | 0.20    |
| When you're driving you need to see the screens, without having to look for them | 0.20    |
| Position of screens not suitable                                                 | 0.29    |
| <b>Incomplete</b>                                                                |         |
| Screen better positioned                                                         | 0.02    |
| Screens are bigger                                                               | 0.19    |
| Screen is too small                                                              | 0.19    |
| <b>Successful</b>                                                                |         |
| Screen seems to be in the continuity of the forms and integrates well            | 0.00    |
| Easy visual change between road and screen                                       | 0.00    |
| Screens placed at a good height above the passenger compartment                  | 0.00    |
| Screens are consistent with the car's design and lines                           | 0.23    |

## 4. Discussion

Analysis is carried out at different levels and information is cross-referenced to give a complete non-expert perception interpretation. Each analysis must be considered as an indicator in order to offer the expert a complete translation support. The perception of three different attributes in nature and interpretation were analyzed. For each of them, information can be provided to expert engineers, whether the attribute is perceived homogeneously or not.



**Fig. 7.** MCA + HAC + K-means results in the first 1x2 factor plane for the Interface Usability attribute.

**Table 5**  
Semantic analysis results for the Interface Usability attribute.

|                                                                                                 | P-value |
|-------------------------------------------------------------------------------------------------|---------|
| <b>Complex</b>                                                                                  |         |
| I get lost between the buttons                                                                  | 0.03    |
| Many small physical buttons aligned that are the same                                           | 0.03    |
| On the screen several menu levels                                                               | 0.03    |
| The screen that takes up the whole dashboard                                                    | 0.08    |
| While driving, you don't get quick information                                                  | 0.08    |
| Tactile buttons are not cut out                                                                 | 0.23    |
| <b>Intermediate</b>                                                                             |         |
| Fewer buttons than usual                                                                        | 0.01    |
| Easier to identify information                                                                  | 0.01    |
| Good balance between screens and the number of buttons                                          | 0.02    |
| It looks like all other vehicles                                                                | 0.07    |
| <b>Simple</b>                                                                                   |         |
| There are screens but there are the necessary buttons to directly access the different features | 0.00    |
| Clear and well-arranged information on the screen                                               | 0.01    |
| Intuitive buttons and pictograms                                                                | 0.03    |
| A suitable screen size                                                                          | 0.24    |
| Buttons as if they were not digital                                                             | 0.24    |

This way of analyzing the data, allows the following conclusions. Visual Load is an attribute that is globally unambiguous in its understanding because there is a consistency within the responses. The consistency is identified at the level of the stimuli classification, at the qualification of the clusters and at the verbalization of categories. All the associated indicators show that there is a homogeneous understanding of the concept behind the attribute. The global consistency found for the Visual Load attribute was expected as it was originally chosen for its ease of understanding (see Table 1). The clear gradient from the most purified to the most loaded stimuli shows that the order relationship induced by the three imposed levels is understood by the non-expert. What is invaluable to the expert engineer when faced with a homogeneous attribute in his perception, is the information that can be drawn from the slight ambiguities. Indeed, it was also expected that the Intermediate level would be equidistant from the Load and Pure ones. But it seems that the distinction is less clear between the Load and the Intermediate levels. From the analysis of the photos, it appears that the unambiguous products are in agreement with the description given in Table 1. While the ambiguous products are surprising: 213 and 457 are loaded with HMI, both should have been seen as charged, whereas 835 and 160 are less loaded with HMI, they should have been seen as less loaded than the others. By analyzing the verbalizations data, we obtain a semantic description of each level which adds weight to the visual analysis and helps to explain the perceptual ambiguity for products located at the edge of clusters. It appears that the dashboard influences non-expert

perception. This is a supplementary argument to support the choice of the sensory evaluation methodology used, because a lot of different dashboards could be presented.

This way of analyzing the data also gives information for an attribute that is globally ambiguous in its understanding, which is invaluable for expert engineers. Indeed, screen Integration is a more difficult attribute to assess, perhaps because of the categorization proposals that are more hedonic in nature (integration well or poorly done). Aesthetic perspective, which is the expert's translation of Screen Integration (see Table 1), seems to be clearly understood by participants according to the semantic analysis. The sentence "screen seems to be in the continuity of the forms and integrates well" is frequently used to characterize well integrated interfaces by non-experts. However, it seems that other criteria could explain this classification: size and screen position relative to the driving. Thus, two main dimensions were used by the participants to classify the stimuli: the aesthetic perspective (visual coherence between screen and dashboard), and the ergonomic perspective (visibility of the screen while driving). This multidimensional interpretation may be the reason for the lack of homogeneity perception. This also indicates that the expert perception is different from the non-expert one. Again, this is a supplementary argument to support the choice of this sensory methodology because results provide access to an understanding of a multidimensional perception and confirms the fact that the hedonic dimension cannot be dissociated from its ergonomic one.

Again, this way of analyzing the data is also useful when stimuli perception is the same among participants, but interpretations are different. Indeed, for the Interface Usability attribute, it has been observed that the stimuli classification was consensual but that the given qualifications of the clusters were not. In this context, it is invaluable for the expert to have an explanation on this identified ambiguity. Stimuli classification analysis shows that the third cluster provokes ambiguity because of their triple level of qualifications (Simple, Intermediate and Complex projection to use at the same time). However, that cluster is the furthest away from the other two, and therefore, in theory, the one whose concept seems the most obvious. The visual descriptive analysis of the representative stimuli, associated with the semantic analysis, helps to get an interpretation. Photos show that vehicles from that cluster are unusual compared to others (very large screen area compared to the cockpit). According to the semantic analysis, concepts translation seems to be more consistent with vehicles from the other two clusters (Simple/Intermediate cluster and Complex/Intermediate cluster). According to participants, what characterizes an HMI in the projection of simple use is the right balance between buttons and screens because access to the main functions is essential. The access can be done by physical buttons or by tactile buttons that are immediately accessible and whose representation evokes the physical ("button as if it were not digital"). However, the visual load (whatever the HMI considered) tends to make the projection in the interaction more complex. Moreover, the display and its organization are important for understanding. A screen with many sub-menus may appear more complex to use and simple icons may make it easier. Thus, the non-expert translation of this attribute does not seem to differ from the expert translation of so-called classic cockpits (see Table 1). However, for less classical cockpits (i.e. with very large screens) the projection in use is maybe more related to the participant's sensitivity or habit of use. Again, this is a supplementary argument to support the choice of this sensory methodology because participants were able to project themselves in an interaction even though only photos were presented to them.

## 5. Conclusion

Some limits were observed and should be considered for future experimentations. For example, this study does not take into account external factors relating to the non-expert, such as the appreciation of technologies. These factors could explain differences in perception, especially concerning more ergonomic attributes. Furthermore, remote

evaluation, without real interaction with interfaces, requires a more advanced description of the HMI for unconventional cockpits that could lead to misunderstanding. Finally, virtual stimuli standardization is not an easy part and should be done by professionals. Indeed, some brands were recognized on the photos by the participants even if this does not seem to have impacted the evaluation process.

Nevertheless, several conclusions can be drawn from this study. Automotive HMI is a complex product to design, to perceive and to evaluate mainly because of its multidimensional nature but also because it is part of a context that cannot be ignored in the understanding of perception. Sensory evaluation, historically developed for the food industry, could be adapted to such complex products, and allows the understanding of their perception. Therefore, a sensory sorting evaluation was set up to quantify and qualify non-expert perceptions on a large number of stimuli. This made it possible to take into account the contextual complexity of the product by presenting HMI embedded at different dashboards. A layer of complexity was added by conducting this evaluation remotely, on photos. It was conclusive even for attributes requiring more interactions. According to the results, the multidimensionality of the product is perceived by the non-expert. Indeed, it seems that the hedonic or sensory dimensions cannot be dissociated from the ergonomic one. In this paper, a methodology for the statistical analysis of the data collected is proposed. As the non-expert perception is invaluable for experts involved in the automotive UX research and development, the process to obtain a visual and textual translation of the non-expert's perception is explained.

Finally, this approach may qualify a semi-structured qualitative interview to the extent that verbalizations data have been collected and analyzed on various pre-defined concepts. But this qualitative basis was reinforced by a quantitative approach. More people than a classical qualitative interview was interviewed, and a sorting task was performed to enhance results. A lot of information can be provided by this semi quantitative–qualitative evaluation. In the light of this experience, this exploratory approach seems promising for a start in co-design.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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